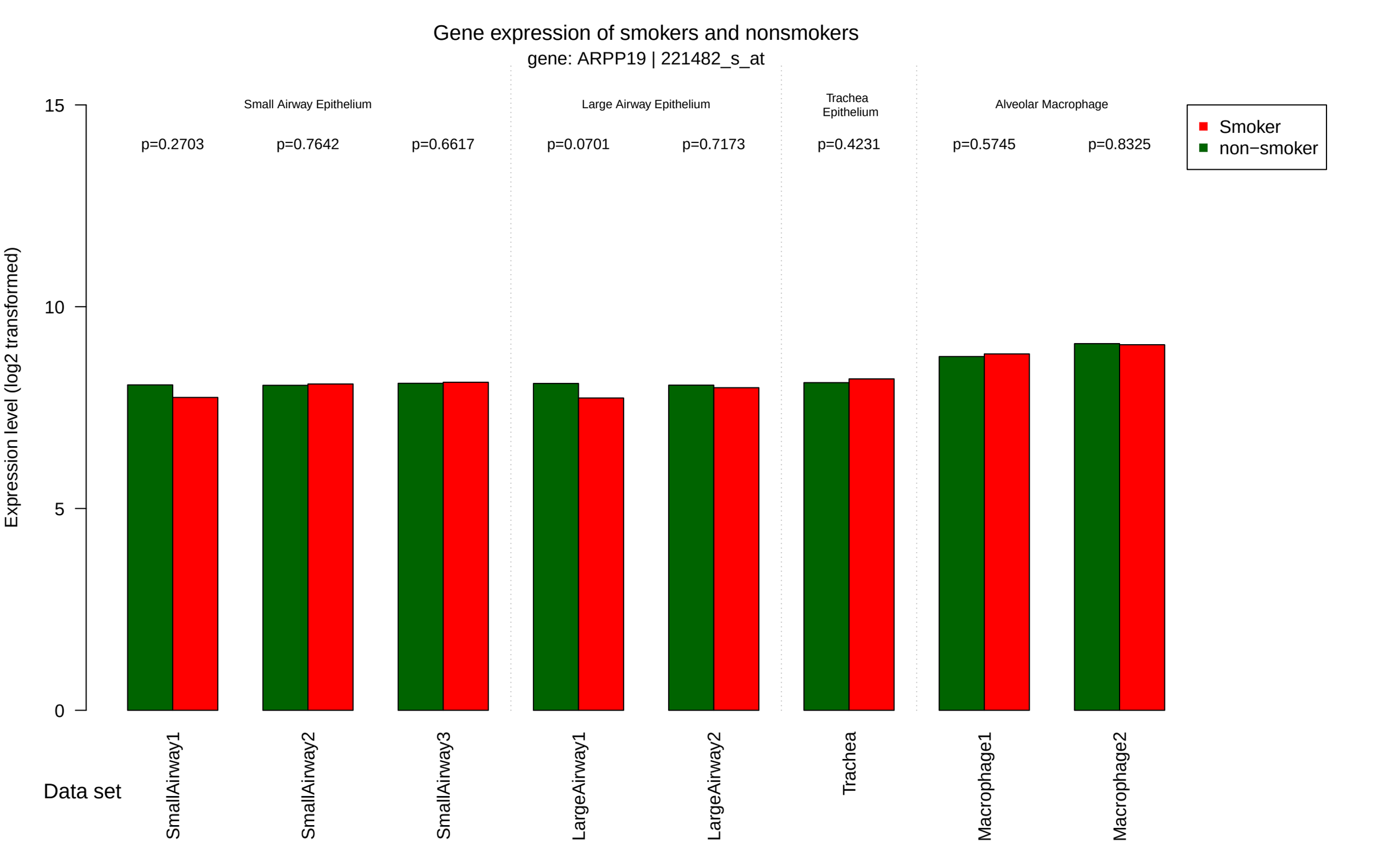
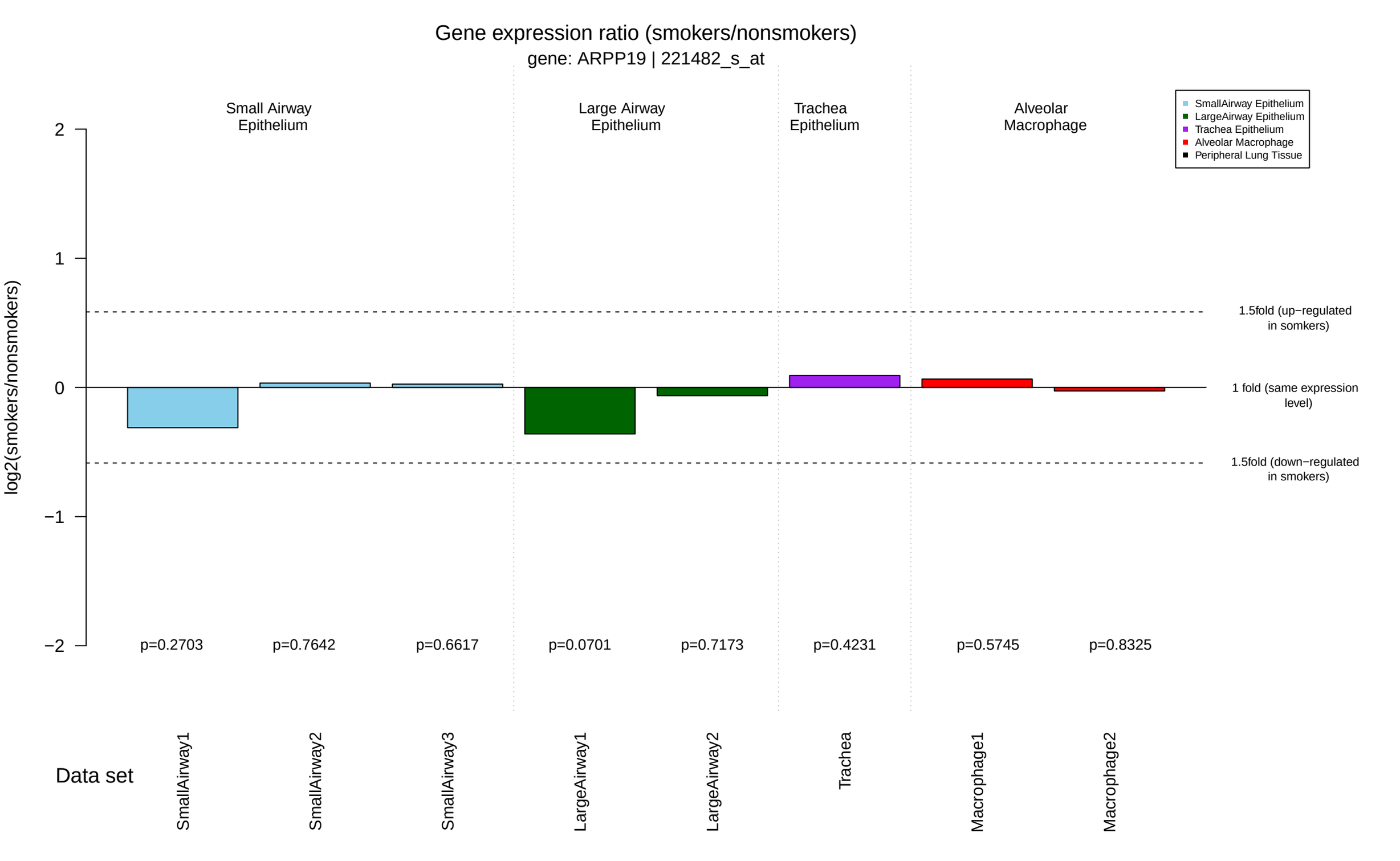
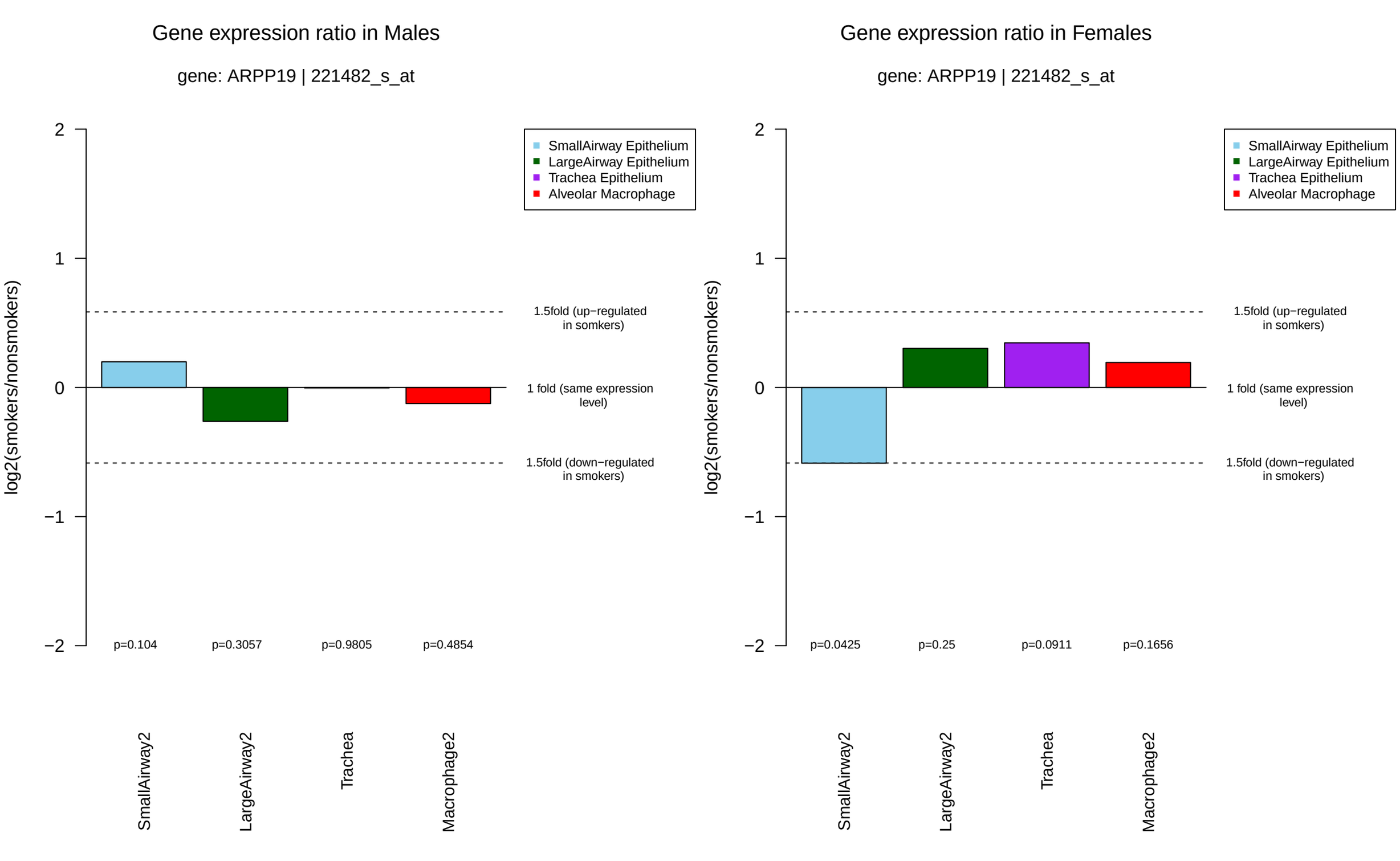
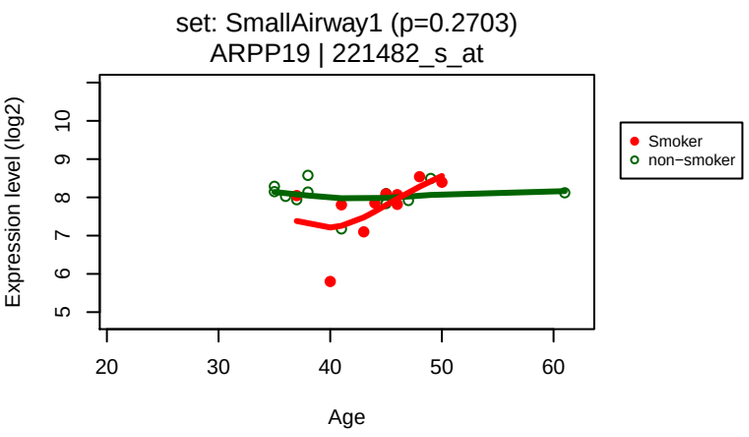
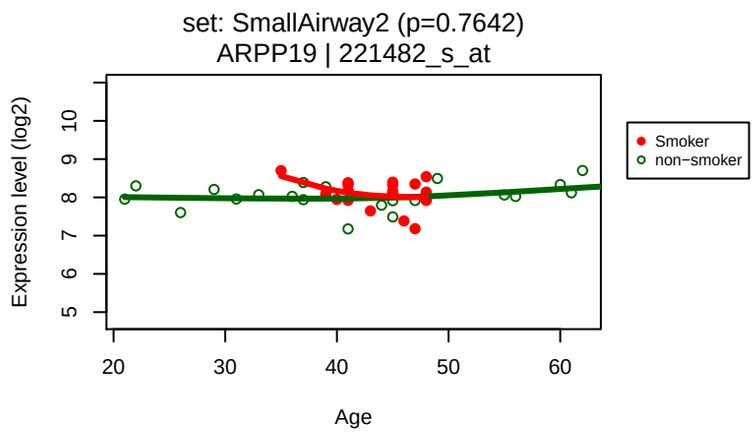
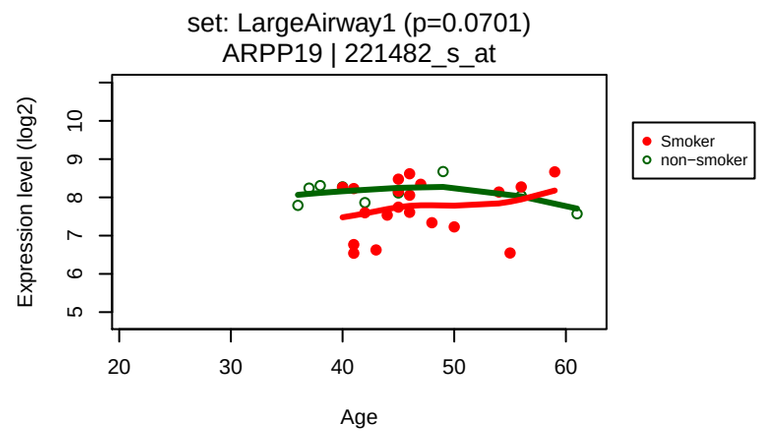

Gene expression of smokers and nonsmokers
gene: ARPP19 | 221482_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2703
p=0.7642
p=0.6617
p=0.0701
p=0.7173
p=0.4231
p=0.5745
p=0.8325
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

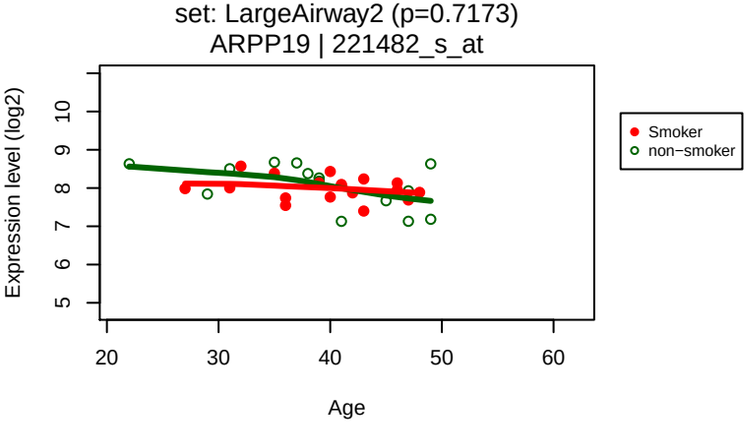

Gene expression ratio (smokers/nonsmokers)
gene: ARPP19 | 221482_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2703
p=0.7642
p=0.6617
p=0.0701
p=0.7173
p=0.4231
p=0.5745
p=0.8325
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

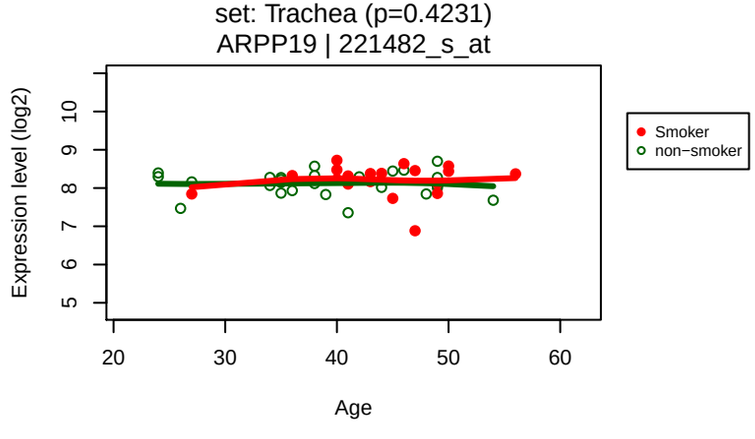

Gene expression ratio in Males
gene: ARPP19 | 221482_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.104
p=0.3057
p=0.9805
p=0.4854
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARPP19 | 221482_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0425
p=0.25
p=0.0911
p=0.1656
SmallAirway2
LargeAirway2
Trachea
Macrophage2

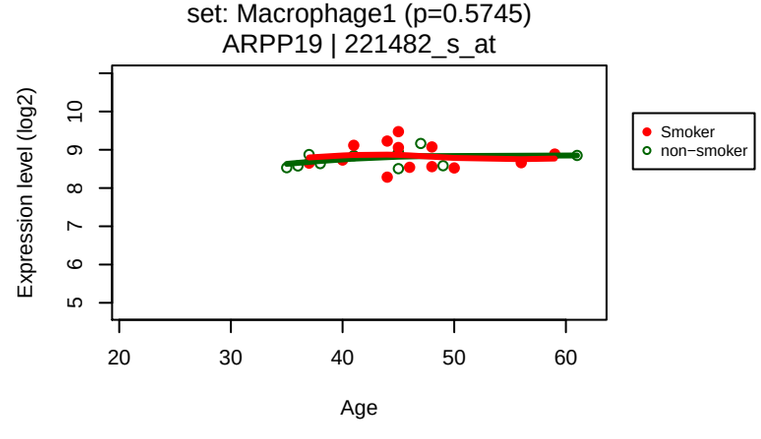

set: SmallAirway1 (p=0.2703)
ARPP19 | 221482_s_at
Expression level (log2)
Age
Smoker
non−smoker

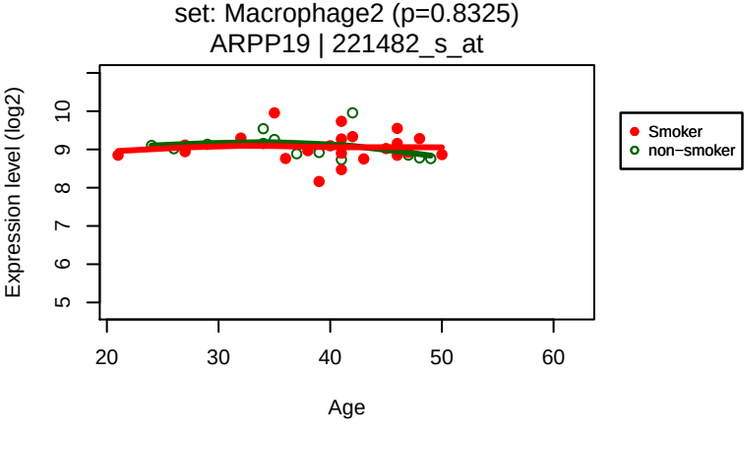

set: SmallAirway2 (p=0.7642)
ARPP19 | 221482_s_at
Expression level (log2)
Age
Smoker
non−smoker

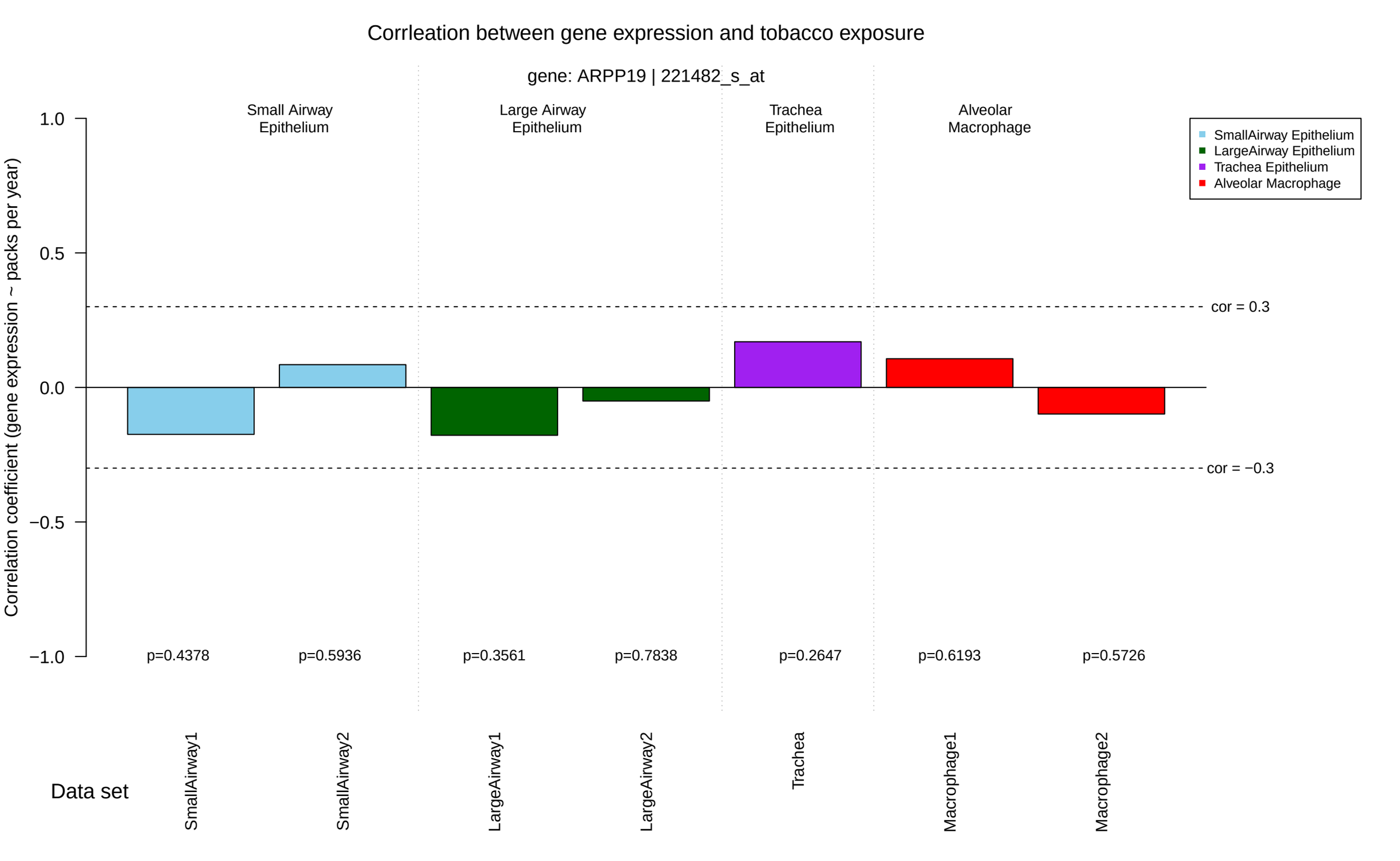

set: LargeAirway1 (p=0.0701)
ARPP19 | 221482_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7173)
ARPP19 | 221482_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4231)
ARPP19 | 221482_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5745)
ARPP19 | 221482_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8325)
ARPP19 | 221482_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARPP19 | 221482_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4378
p=0.5936
p=0.3561
p=0.7838
p=0.2647
p=0.6193
p=0.5726
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2